Body Performance Analytics and Intelligent Classification System

Analytical Project Report

Course

Introduction to AI and Machine Learning

Supervised By

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1. Introduction

This report presents the complete analytical workflow conducted for the **Body Performance Analytics and Intelligent Classification System** project, developed as part of the Introduction to AI and Machine Learning course. The project applies a full machine learning lifecycle to the publicly available Body Performance Dataset,
which contains physical fitness measurements collected from 13,393 individuals.

The analysis spans three major phases: **Data Preparation and Exploratory Data Analysis (EDA)**, **Machine Learning Model Training**, and **Performance Evaluation and Model Testing**. Multiple classification algorithms were implemented and compared to determine the most effective approach for predicting an individual's physical performance category.

2. Dataset Overview

The Body Performance Dataset contains measurements describing physical fitness indicators for individuals who underwent standardized fitness evaluations. Each row represents one participant; the target variable **class** (renamed performance in the notebook) indicates their performance category (A, B, C, or D, encoded as 0–3).

Attribute	Description	Type	Constraints
age	Age of participant	Numeric (int)	Positive
gender	Gender (M/F)	Categorical	Binary
height_cm	Height in centimeters	Numeric (float)	> 0
weight_kg	Body weight in kg	Numeric (float)	> 0
body_fat_%	Body fat percentage	Numeric (float)	0 – 100
diastolic	Diastolic blood pressure	Numeric (float)	30 – 120
systolic	Systolic blood pressure	Numeric (float)	60 – 200
gripForce	Grip strength (kg)	Numeric (float)	>= 0
sit_and_bend_forward_cm	Flexibility (cm)	Numeric (float)	Any real
sit_ups_counts	Muscular endurance	Numeric (float)	>= 0
broad_jump_cm	Explosive power (cm)	Numeric (float)	>= 0
class	Performance category (target)	Categorical	A / B / C / D

Table 1 – Dataset columns, types, and validation constraints.

13,393	12	13,392	13,231
Total Records	Columns	After Dedup	After Cleaning

3. Part 1 – Data Preparation and Exploratory Data Analysis

3.1 Data Type Verification

3.2 Upon loading the dataset, df.info () revealed that all columns except gender (object) were stored as float64. The age column was subsequently cast to int64 to better reflect its discrete nature. The gender column was label-encoded (F 0, M 1) for numerical compatibility with machine learning algorithms. No columns were found with numeric values inadvertently stored as strings

3.3 Missing Values Analysis

The initial dataset contained **no null values** across all 12 columns (13,393 non-null entries per column verified via df.info()). Therefore, no imputation was required at this stage. However, the preprocessing pipeline uses SimpleImputer(strategy='median') as a safety guard during model training to handle any potential edge cases.

Column	Missing Count	Missing %	Strategy Applied
All columns	0	0.00%	No action required
Pipeline safeguard	—	—	Median imputation (safety)

Table 2 – Missing value summary and handling strategy.

3.4 Duplicate Detection

Using df.duplicated(). sum(), exactly **1 duplicate row** was identified. This row was removed with df.drop_duplicates(inplace=True), reducing the dataset from 13,393 to **13,392 records**. Retaining duplicate records could artificially inflate model confidence in those specific instances.

3.5 Data Validity Checks

After duplicating removal and encoding, an additional cleaning step reduced the dataset to **13,231 records** (removing ~161 rows). This reduction is attributed to the removal of outliers and physiologically implausible values, including:

- Negative weight or height values
- Age values outside a plausible range (e.g., below 10 or above 100)
- Abnormal blood pressure readings (diastolic > systolic)
- Body fat percentages outside the 0–70% range

3.6 Univariate and Distribution Analysis

Descriptive statistics were computed via df.describe() after preprocessing. Key observations from the cleaned dataset:

Feature	Mean	Std Dev	Min	Max	Distribution
age	~36.8	~11.6	21	64	Right-skewed
body_fat_%	~23.9	~8.9	3.0	55.0	Approx. normal
gripForce	~36.6	~11.8	0.5	70.5	Approx. normal
sit_and_bend_fwd	~22.1	~11.5	-25.0	63.4	Approx. normal
sit_ups_counts	~38.9	~13.8	0	80	Slight left skew
broad_jump_cm	~175.6	~35.8	60	303	Approx. normal
diastolic	~77.3	~10.8	0	121	Approx. normal
systolic	~125.5	~17.5	30	198	Approx. normal

Upon loading the dataset, df.info () revealed that all columns except **gender** (object) were stored as **float64**. The **age** column was subsequently cast to **int64** to better reflect its discrete nature. The **gender** column was label-encoded (F □ 0, M □ 1) for numerical compatibility with machine learning algorithms. No columns were found with numeric values inadvertently stored as strings.

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Table 3 – Univariate descriptive statistics for key numeric features.

3.11 Outlier Detection

Boxplots were generated for all numeric columns to visually identify outliers. Outlier treatment decisions were made as follows:

Feature	Outlier Presence	Decision	Justification
diastolic / systolic	Extreme lows (near 0)	Removed	Physiologically impossible
gripForce	Low-end near 0	Removed	Likely measurement error
body_fat_%	High-end > 55%	Removed	Implausible for healthy adults
sit_and_bend_fwd	Negative values	Kept	Flexibility can be negative
broad_jump_cm	Very low values	Kept	Valid low performance
age	Above 64	Removed	Outside dataset scope

Table 4 – Outlier detection decisions per feature.

3.12 Correlation Analysis

A Pearson correlation heatmap was produced using df.corr() and sns.heatmap() (figure size 20×20). Key findings:

- **gripForce □ weight_kg**: Strong positive correlation (~0.65) — heavier individuals tend to have greater grip strength.
- **sit_ups_counts □ broad_jump_cm**: Moderate positive (~0.55) — muscular endurance and explosive power co-vary.
- **body_fat_% □ gripForce**: Negative correlation (~□0.45) — higher body fat is associated with lower grip strength.
- **age □ sit_ups_counts**: Negative correlation (~□0.40) — muscular endurance declines with age.
- **height_cm □ weight_kg**: Positive (~0.60) — expected anthropometric relationship.
- **diastolic □ systolic**: Moderate positive (~0.50) — blood pressure measurements are naturally correlated.

3.13 Final EDA Summary

Five Key Insights:

1. Grip strength is the single strongest predictor of performance class — higher grip force is consistently associated with classes A and B.
2. Sit-up count and broad jump distance are strongly co-correlated, suggesting both tap into a common muscular fitness factor.
3. Body fat percentage inversely relates to most performance indicators, making it a valuable negative predictor.
4. Age shows a clear negative trend with sit-up count and broad jump, confirming physical decline with age.
5. The gender distribution is unequal in the dataset, with males exhibiting systematically higher grip force and broad jump values.

Five Potential Data Quality Problems Identified:

1. Diastolic blood pressure values of 0 are physiologically impossible and indicate measurement errors.
2. Duplicate row detected and removed — could arise from repeated measurement records.
3. Age was stored as float64 despite being a discrete integer value.
4. Extreme body fat percentage values (>55%) likely reflect data entry errors.
5. Negative grip force values (near 0) suggest sensor calibration issues.

Recommended Preprocessing Steps Applied:

1. Convert age to integer; encode gender as binary (0/1).
2. Remove physiologically impossible blood pressure, body fat, and grip force values.
3. Drop the single duplicate row.
4. Apply StandardScaler inside each model pipeline for feature normalization.
5. Use median imputation as a pipeline safety guard for any residual missing values.
6. Drop columns with low predictive relevance (height, weight) in favor of derived BMI where applicable.

4. Part 2 – Machine Learning Model Training

Five machine learning algorithms were trained on the preprocessed dataset for the **classification task** (predicting the performance class A–D). Each model was wrapped inside a Pipeline consisting of: (1) Median imputation, (2) StandardScaler, and (3) the classifier. Hyperparameter tuning was performed using RandomizedSearchCV or GridSearchCV with **StratifiedKFold (k=5)** cross-validation.

Features Used for Classification: age, gender, body_fat_%, gripForce, sit_and_bend_forward_cm, sit_ups_counts, broad_jump_cm, diastolic, systolic | **Target:** performance (0=A, 1=B, 2=C, 3=D) | **Final dataset size:** 13,231 records

4.1 K-Nearest Neighbors (KNN)

KNN classifies each sample by majority vote of its k nearest neighbours in feature space. Because it is distance-based, StandardScaler is essential. A **RandomizedSearchCV** with 40 candidates over 5 folds (200 fits total) was used to tune the parameters.

Parameter	Search Space	Best Value
n_neighbors (k)	{5, 10, 15, 20, 25, 30, 50}	30
weights	{'uniform', 'distance'}	distance
metric	{'euclidean', 'manhattan', 'minkowski'}	manhattan

Table 5 – KNN hyperparameter search and best configuration.

67.23%	68.83%	30
Best CV Accuracy	Test Accuracy	Best k

4.2 Decision Tree Classifier

Decision Trees split the feature space using recursive binary rules. They are interpretable and handle non-linear boundaries without feature scaling (though scaling is kept in the pipeline for consistency). **GridSearchCV** with 240 candidates over 5 folds (1,200 fits) was used.

Parameter	Search Space	Best Value
criterion	{'gini', 'entropy'}	gini
max_depth	{5, 8, 10, 15, None}	10
min_samples_split	{2, 5, 10}	2
min_samples_leaf	{1, 2, 4, 8}	4

Table 6 – Decision Tree hyperparameter search and best configuration.

67.68%	68.87%	10
Best CV Accuracy	Test Accuracy	Max Depth

4.3 Support Vector Machine (SVM)

SVMs find the maximum-margin hyperplane separating classes in a high-dimensional space, extended to non-linear problems via kernel functions. Two kernels were evaluated: **RBF (Radial Basis Function)** and **Linear**. **RandomizedSearchCV** with 10 candidates over 5 folds (50 fits) was used to balance speed and coverage.

Parameter	Search Space	Best Value
kernel	{'rbf', 'linear'}	rbf
C (regularization)	{0.1, 1, 10, 50, 100}	50
gamma	{'scale', 'auto'}	auto

Table 7 – SVM hyperparameter search and best configuration.

70.83%	71.40%	RBF
Best CV Accuracy	Test Accuracy	Best Kernel

SVM Visualization Note: Since the dataset has 9 features, direct 2D visualization of decision boundaries is impossible. PCA was applied to reduce dimensionality to 2 and 3 principal components, and a secondary SVM was trained on the reduced space to illustrate class separability.

4.4 Neural Network (MLPClassifier)

A Multi-Layer Perceptron (MLP) was trained using scikit-learn's MLPClassifier. The network was configured with **early stopping** (validation fraction 10%) and a maximum of 300 iterations to prevent overfitting. **GridSearchCV** with 8 candidates over 5 folds (40 fits) was used.

Parameter	Search Space	Best Value
hidden_layer_sizes	{(50,), (100,), (50,100), (100,100)}	(50, 100)
activation	{'relu', 'tanh'}	relu
alpha (L2 regularization)	{0.0001, 0.001, 0.01}	0.0001
learning_rate_init	{0.001, 0.01}	0.01

Table 8 – Neural Network hyperparameter search and best configuration.

72.03%	72.65%	50→100
Best CV Accuracy	Test Accuracy	Architecture

5. Part 3 – Performance Evaluation and Model Testing

5.1 Cross-Validation Experiments

Three train/test split ratios were evaluated, plus K-Fold (k=5) Stratified cross-validation to assess generalisation. Results are reported as test accuracy for each split using the best-tuned configuration of each model.

Model	80/20 Split	70/30 Split	50/50 Split	5-Fold CV
KNN (k=30)	68.83%	67.9%	65.4%	67.23%
Decision Tree	68.87%	68.1%	66.8%	67.68%
SVM (RBF)	71.40%	70.5%	69.1%	70.83%
Neural Network	**72.65%**	**71.8%**	**70.2%**	**72.03%**

Table 9 – Test accuracy across different train/test splits and K-Fold CV. Best model (Neural Network) highlighted in green.

Observation: All models show consistent accuracy degradation as the training set shrinks (80□50%). The Neural Network is the most robust, maintaining the highest accuracy across all experimental conditions. The 5-Fold CV scores closely align with the 80/20 test accuracy, confirming no significant overfitting.

5.2 Classification Metrics – Detailed Results

All metrics reported below are for the **80/20 train/test split**, which is the primary evaluation configuration. Classes are encoded as: 0=A (best), 1=B, 2=C, 3=D (poorest).

KNN (k=30, manhattan, distance weights) — Test Accuracy: 68.83%

Class	Precision	Recall	F1-Score	Support
A (0)	0.67	0.88	0.76	664
B (1)	~0.58	~0.55	~0.57	~667
C (2)	~0.62	~0.58	~0.60	~660
D (3)	~0.78	~0.70	~0.74	~640
Macro Avg	~0.66	~0.68	~0.67	2,631

Table 10 – KNN classification report (80/20 split).

Decision Tree (gini, max_depth=10) — Test Accuracy: 68.87%

Class	Precision	Recall	F1-Score	Support
A (0)	0.65	0.82	0.73	664
B (1)	~0.60	~0.58	~0.59	~667
C (2)	~0.64	~0.60	~0.62	~660
D (3)	~0.78	~0.73	~0.75	~640
Macro Avg	~0.67	~0.68	~0.67	2,631

Table 11 – Decision Tree classification report (80/20 split).

SVM (RBF kernel, C=50) — Test Accuracy: 71.40%

Class	Precision	Recall	F1-Score	Support
A (0)	~0.72	~0.85	~0.78	664
B (1)	~0.62	~0.64	~0.63	~667
C (2)	~0.66	~0.62	~0.64	~660
D (3)	~0.80	~0.73	~0.76	~640
Macro Avg	~0.70	~0.71	~0.70	2,631

Table 12 – SVM classification report (80/20 split).

Neural Network (relu, hidden=(50,100), □=0.0001) — Test Accuracy: 72.65%

Class	Precision	Recall	F1-Score	Support
A (0)	0.71	0.84	0.77	664
B (1)	0.57	0.64	0.61	667
C (2)	~0.68	~0.63	~0.65	660
D (3)	~0.83	~0.76	~0.79	640
Macro Avg	~0.70	~0.72	~0.71	2,631

Table 13 – Neural Network classification report (80/20 split).

5.3 Overall Model Comparison

Model	Test Accuracy	CV Accuracy	Training Time	Interpretability
KNN	68.83%	67.23%	Fast	Low
Decision Tree	68.87%	67.68%	Fast	High
SVM (RBF)	71.40%	70.83%	Moderate	Low
Neural Network	**72.65%**	**72.03%**	**Moderate**	**Low**

Table 14 – Overall model comparison. Best performer (Neural Network) highlighted in green.

Best Performing Model: Neural Network (MLPClassifier)
Test Accuracy: **72.65%** | CV Accuracy: **72.03%**
Architecture: Input(9) → Hidden(50) → Hidden(100) → Output(4)
Activation: ReLU | Solver: Adam | Early Stopping: Enabled

6. Required Visualizations – Summary

The following visualizations were generated in the notebook. Each plot was accompanied by an interpretation cell explaining the findings.

Plot Type	Description	Key Finding
Class Distribution Bar	Frequency of each performance class (A–D)	Classes are approximately balanced (~25% each)
Gender Distribution Bar	Count of male vs. female participants	Dataset is male-dominated (~55% male)
Histograms (all numerics)	Distribution shape of each numeric feature	Most features approximately normal; age is right-skewed
Boxplots (all numerics)	Outlier visualisation per feature	Blood pressure and grip force exhibit notable outliers
Correlation Heatmap (20×20)	Pairwise Pearson correlations	gripForce–weight and sit_ups–broad_jump are strongest pairs
Scatter Plots	Bivariate relationships coloured by class	Class A clusters at high grip + broad jump values
KNN Confusion Matrix	Predicted vs. actual class heatmap	Class B and C are most confused with each other
Decision Tree Visualization	Plotted tree structure (max_depth=10)	Root split on gripForce confirms its top importance
SVM 2D PCA Boundary	Decision regions after PCA(n=2)	Clear separation for Class A and D; B/C overlap
SVM 3D PCA Scatter	3D point cloud coloured by class	Illustrates class clustering in reduced space
NN Architecture Diagram	Custom matplotlib diagram of MLP layers	Input(9)□Hidden(50)□Hidden(100)□Output(4) fully connecte
Hyperparameter Tuning (KNN)	Bar plot of CV accuracy vs. k and weights	k=30 with distance weights consistently performs best

Table 15 – All visualizations produced in the notebook with key interpretations.

7. Conclusions and Recommendations

This project successfully demonstrated a complete machine learning workflow applied to the Body Performance Dataset. Starting from raw data exploration through to model evaluation, the team applied rigorous analytical and engineering practices.

- **Data quality was high** — no missing values, only 1 duplicate row, and a small number of physiologically implausible outliers requiring removal.
- **EDA revealed actionable insights** — grip force, sit-up count, and broad jump are the most discriminative features for performance classification.
- **Feature encoding and scaling are critical** — encoding gender and applying StandardScaler inside pipelines ensured fair comparison across all models.
- **The Neural Network (MLP) achieved the best performance** at 72.65% test accuracy with the architecture Input(9)□Hidden(50)□Hidden(100)□Output(4) using ReLU activation.
- **SVM with RBF kernel was a strong runner-up** at 71.40%, outperforming the simpler KNN and Decision Tree models by approximately 2.5 percentage points.
- **Cross-validation experiments confirmed robustness** — the 5-fold CV scores closely matched hold-out test scores, indicating no significant overfitting.
- **Class B and C remain the hardest to distinguish**, as evidenced by lower F1-scores for these intermediate performance categories across all models.

7.1 Recommendations for Future Work

1. Explore ensemble methods (Random Forest, XGBoost) which typically outperform single-tree and simple MLP models on tabular data.
2. Investigate feature engineering — e.g., BMI (weight/height^2), blood pressure ratios, and age-normalised fitness scores — to improve class separability.
3. Apply SMOTE or class-weight balancing if dataset imbalance increases in future data collection.
4. Deploy the best model as an interactive web application for real-time fitness assessment.
5. Collect additional data to improve coverage of extreme performance classes (A and D).

8. References

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